FORMAL SYSTEMS DEPEND ONLY ON THE SUPPORT: FOR X NOETHERIAN AND I, J QUASI-COHERENT SHEAVES OF IDEALS WITH THE SAME VANISHING CLOSED SUBSET, THE CATEGORIES COH(X, I) AND COH(X, J) ARE EQUIVALENT.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6922-6930 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 6932-6960, the complete author proof (27 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool stacks-coherent backfill wave 33d. precard_gate RESULT=OK (probe L9999).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 00IM: sources/stacks-coherent/pinned-original-source.tex, source lines 6960–6960. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0319: sources/stacks-coherent/pinned-original-source.tex, source lines 6960–6960. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00DY: sources/stacks-coherent/pinned-original-source.tex, source lines 6960–6960. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087W: sources/stacks-coherent/pinned-original-source.tex, source lines 6489–6495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a Noetherian scheme. Let $\mathcal{I}, \mathcal{J} \subset \mathcal{O}_X$ be quasi-coherent sheaves of ideals. If $V(\mathcal{I}) = V(\mathcal{J})$ is the same closed subset of X, then $Coh(X, \mathcal{I})$ and $Coh(X, \mathcal{J})$ are equivalent.*

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. First, assume $X = \mathrm{Spec}(A)$ is affine. Let $I, J \subset A$ be the ideals corresponding to $\mathcal{I}, \mathcal{J}$. Then $V(I) = V(J)$ implies we have $I^c \subset J$ and $J^d \subset I$ for some $c, d \geq 1$ by elementary properties of the Zariski topology (see Algebra, Section 00DY and Lemma 00IM). Hence the I-adic and J-adic completions of A agree, see Algebra, Lemma 0319. Thus the equivalence follows from Lemma 087W in this case.

In general, using what we said above and the fact that X is quasi-compact, to choose $c, d \geq 1$ such that $\mathcal{I}^c \subset \mathcal{J}$ and $\mathcal{J}^d \subset \mathcal{I}$. Then given an object $(\mathcal{F}_n)$ in $Coh(X, \mathcal{I})$ we claim that the inverse system

$$(\mathcal{F}_{cn}/\mathcal{J}^n\mathcal{F}_{cn})$$

is in $Coh(X, \mathcal{J})$. This may be checked on the members of an affine covering; we omit the details. In the same manner we can construct an object of $Coh(X, \mathcal{I})$ starting with an object of $Coh(X, \mathcal{J})$. We omit the verification that these constructions define mutually quasi-inverse functors. $\square$